# LAN & IM Communication API Guide

## Mobile App

### CommChannelManager class

```
/**
 * Initiate
 * @param context
 */
public static void init(Context context);


/**
 * Create Channel
 * @param channelName
 * @return
 */
public Channel createChannel(String channelName);


/**
 * Get Channel by Name
 * @return
 */
public Channel getChannelByName(String channelName);

/**
 * Get Connections by Robot SN
 * @return
 */
public ArrayList<Connection> getConnectionsByRobotSn(String sn);

/**
 * Get Connections by Robot Channel Name
 * @return
 */
public ArrayList<Connection> getConnectionsByChannelName(String channelName);


public RobotMaster createRobotMaster(Connection connection);
```

RobotMaster Claee

```
/**
 * Subscribe Event
 * @param subscriber
 * @param actions
 */
void subscribe(EventSubscriber subscriber, String action，  ResultCallback callback);

/**
 * Unsubscribe Event
 * @param subscriber
 */
void unsubscribe(EventSubscriber subscriber，  ResultCallback callback);

/**
 * Create Robot Service
 *
 * @param serviceName
 * @return Service Proxy
 */
RobotServiceProxy createRobotSystemServiceProxy(String serviceName)

/**
 * Get Robot Skill Proxy
 * @return
 */
RobotSkillsProxy createRobotSkillsProxy();
```

RobotService and RobotApplicationCluster implement ConvenientCallable

```
void setConfiguration(CallConfiguration configuration);

CallConfiguration getConfiguration();

Response call(String path) throws CallException;

Response call(String path, @Nullable Param param) throws CallException;

Cancelable call(String path, @Nullable ResponseCallback callback);
```

```
Cancelable call(String path, @Nullable Param param, @Nullable ResponseCallback callback);
```

Channel Class

```
/**
 *Create IM Connection
 * @return IM Connection
 */
public ImConnection createImConnection();

/**
 * Create LAN Connection
 * @return LAN Connection
 */
public SocketConnection createSocketConnection() ;
```

ImConnection and SocketConnection public interface (Connection class)

```
/**
 * connect
 * @param parameter LanConnectParam for LAN connection, ImConnectParam for IM conenction
 * @param callback
 */
void connect(Parameter parameter, @Nullable ConnectionCallback callback);

/**
 * Disconnect
 * @param callback
 */
void disconnect(@Nullable ConnectionCallback callback);

/**
 * Get Connection State
 * @return
 */
INet.ConnectionState getState();

/**
 * Release resources
 */
void release();
```

```
/**
 * Register Connection State Listener
 * @param listener
 */
void registerConnectionStateListener(@Nullable ConnectionStateListener listener);

/**
 * Unregister Connection State Listener
 * @param listener
 */
void unregisterConnectionStateListener(@Nullable ConnectionStateListener listener);

/**
 * Add Push Message Listener
 * @param listener
 */
void addPushMessageListener(PushMessageListener listener);

/**
 * Remove Push Message Listener
 * @param listener
 */
void removePushMessageListener(PushMessageListener listener);

/**
 * Whether connection is available
 * @return
 */
boolean isAvailable();

/**
 * Get Connection Type
 * @return     SOCKET_TYPE | IM_TYPE
 */
ConnectionType getConnectionType();
```

### Differentiated API (Customized according to IM Provider)

IM

```
/**
 * Add Message listener
```

```
     * @param receiveListener
     */
    void addMessageListener(@NonNull MessageListener receiveListener);

    /**
     * Remove Message listener
     * @param receiveListener
     */
    void removeMessageListener(@NonNull MessageListener receiveListener) ;

    /**
    * Get Conversation
    */
    ImConversation getConversation();
```

ImConversation

```
    /**
     * Send Message
     * @param message
     * @param callback
     */
    void sendMessage(ImMessage message, SendCallback callback);

    /**
     * Send File
     * @param path     File path
     * @param fileName File name shown in the receiver
     * @param callback
     */
    void sendFile(String path, String fileName, SendCallback callback);
```

# Robot

CommChannelManager class

```
    /**
     * Initiate
     * @param context
     */
    public static void init(Context context);
```

```
/**
 * Get IM Channel
 * @return IM Channel
 */
public static ImChannelHub getImChannelHub();

/**
 * Get IM Channel
 * @return IM Channel
 */
public static SocketChannelHub getSocketChannelHub();
```

### LanChannelHub

```
/**
 * Get LAN Channel
 * @return LAN Channel
 */
public static ArrayList<SocketConnection> getSocketConnections();
```

### LanConnection

```
/**
* Push Message to Mobile App
* @param msg
*/
public void push(ProtoParam msg, ResultCallback callback);
```

### ImChannelHub

```
/**
 * Add FileMessage Listener
 * @param receiveListener
 */
void addFileMessageListener(@NonNull FileMessageListener receiveListener);

/**
 * Remove FileMessage Listener
 * @param receiveListener
 */
void removeFileMessageListener(@NonNull FileMessageListener receiveListener) ;
```

```
/**
 * Get Conversation
 * @param peer
 */
ImConversation getConversation(String peer);
```

ImConversation

```
/**
 *    Push Message to Mobile App
 * @param msg
 */
 public void push(ProtoParam msg, ResultCallback callback);

/**
    * Sned Message
    * @param message
    * @param callback
    */
  void sendMessage(ImMessage message, SendCallback callback);

   /**
    * Send file
    * @param path     File Path
    * @param fileName File Name shown in the receiver
    * @param callback
    */
  void sendFile(String path, String fileName, SendCallback callback);
```